Liability for the Production and Sale of Vaccines

Potential liability for vaccine-related injuries has received much attention as a deterrent to vaccine manufacturing.

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Experience From the Swine Flu Vaccination Program

These issues received highly publicized national attention in the summer of 1976. The discovery of swine flu virus in a human population with resulting illness (and one death) at Fort Dix, New Jersey, in January 1976, led the CDC to recommend a national program to provide immunizations against swine flu for the population.[6] The need for the program was based on the extensive morbidity and mortality experienced during the disastrous pandemic of 1918, caused by an apparently identical strain of influenza virus.

The proposals for such a vaccination program moved through the executive branch and the President to the Congress without a full appreciation of the problems presented by the

threat of liability. The program was formulated and advanced on the assumption that the vaccine manufacturers would gladly produce for a fee close to cost the needed vaccine. The public health officials and government lawyers did not realize that Reyes and Givens in the district court represented an entirely new factor in the decision to produce and sell vaccines, particularly a vaccine proposed for administration to nearly the entire population in a short period.

The manufacturers and their insurance carriers, on the other hand, no doubt resented the liability burdens placed upon them by Judge Wisdom, and were perhaps not unhappy about the opportunity to have their concerns addressed in a national forum. In any case, it became clear that the program would not go forward without some legal protection for the manufacturers, who were unable to obtain insurance at a price acceptable to them and refused to carry the risk themselves, in large part because of the limited time available for safety testing.

Rapid congressional action necessitated by the need to manufacture and administer the vaccine before the onset of the flu season resulted in P.L. 94-380, effective August 11, 1976. The act provided for an exclusive remedy against the United States "for personal injury or death arising out of the administration of swine flu vaccine

under the swine flu program and based upon the act or omission of a program participant" [P.L. 94-380 sec. 2(k) (2) (A)]. Program participant was defined broadly to include

> . . . the manufacturer or distributor of the swine flu vaccine used in an inoculation under the swine flu program, the public or private agency or organization that provided an inoculation under the swine flu program without charge for such vaccine or its administration and in compliance with the informed consent form and procedures requirements prescribed [in the statute], and the medical and other health personnel who provided or assisted in providing an inoculation under the swine flu program without charge for such vaccine or its administration and in compliance with such informed consent form and procedures requirements.

The statute left open the possibility of the manufacturer's or other participant's liability only in the form of a right of the United States to sue any participant for the amount it had paid and the costs of defending the litigation "resulting from the failure of any program participant to carry out any obligation or responsibility assumed by it under a contract with the United States in connection with the program or from any

negligent conduct on the part of any program participant . . ." [Id. sec. 2(k)(7)].

The Swine Flu Act had little to recommend it except that it made it possible to go ahead with swine flu vaccine production. The manufacturers obtained protection for one vaccine under one temporary program, but no attention to the issues on any systematic basis. The law was criticized from many points of view but, in fact, it only changed the defendant and the rules, albeit unclear, remained the same. It also thrust the government into the unfamiliar role of product liability and malpractice defendant. The preamble to section 2 (k) stated that the statute was necessary "in order to be prepared to meet the potential emergency . . . until Congress develops a permanent approach for handling claims arising under programs of the Public Health Service Act." That "permanent approach" subsequently faded as rapidly as the swine flu emergency.

The concern about the liability problems of vaccination programs so visible in the debate over swine flu carried over to the state level. To further the cause of universal vaccination and to protect participants in the process, the manufacturers, the American Academy of Pediatrics, and others sought state legislation limiting liability. In the wake of the swine flu controversy, they were successful in two states: Maryland and California.

Since then, no new, relevant state legislation has passed.

The Maryland statute provides for immunity of all "program participants," including manufacturers, if the state Secretary of Health and Mental Hygiene finds that an immunization project conforms to good medical and public health practice and gives written approval. No person is exempted from liability for gross negligence, and a manufacturer is not exempted "from the duty to use ordinary care in preparing and handling a drug or vaccine" [Maryland Code sec. 18-401, revised version of Ch. 238 of the 1977 laws].

The California statute mandated the creation of a special fund to provide for the medical, institutional, supportive, and rehabilitative care required "because of severe adverse reaction to any immunization required by state law to be administered to children under 18 years of age."[7] The statute also stated that "No person shall be liable for any injury caused by an act or omission in the administration of a vaccine or other immunizing agent to a minor . . . if such immunization is required by state law and the act or omission does not constitute willful misconduct or gross negligence" [California Health and Safety Code secs. 429.35 and 429.36, passed as Ch. 1097, secs. 1 and 2 of the 1977 statutes]. A California trial court has held that this statute

protects manufacturers from liability [Flood v. Wyeth Laboratories, Inc., No. SW C 58664 (Superior Court for the County of Los Angeles)].

The swine flu program was not a public health success. The vaccine was successfully manufactured, distributed, and administered to over 45 million people in a relatively short period of time—a considerable administrative and logistical achievement. However, the virus did not reappear the following winter, for reasons that no one understands. Furthermore, it appeared that 1 out of 100,000 of the recipients developed Guillain-Barré syndrome (the most recent study reports a rate of 1 per 200,000)[8], an unexpected and unique occurrence of unknown etiology. As a result of the publicity about side effects and the documentation of the apparent relationship between the vaccine and Guillain-Barré syndrome (GBS), a large number of claims were filed. The special liability provisions of the swine flu statute, which very well might have faded into history, proved to be important.

For the purposes of this study, the main importance of the provisions is that they resulted in a large body of litigation relating to the issue of liability for vaccine manufacture and administration. The claims and suits are summarized in Table 6.1, prepared by the Torts Branch of the Civil Division of the Department of

Justice. The government has paid more than $73 million to claimants alleged to have been injured by the swine flu vaccination. Although this amount is large in absolute terms, it is less than $2 per vaccine recipient.

TABLE 6.1

Swine Flu Claims, Suits, and Outcomes (as of December 15, 1983).

Because there has been so little litigation directly relevant to the liability issue for vaccine manufacture and administration, the numerous swine flu cases provide the best guide to date on the legal rights of injured recipients, the liability risks of manufacturers, and the likely results of court action. It is useful, therefore, to describe this experience at some length and to evaluate the factors that make it relevant to the general question of what vaccine manufacturers need or need not fear.

The most important decision made in connection with the swine flu litigation was a decision by the government to stipulate to liability (accept liability responsibility) for damages caused by the GBS in any case where the plaintiff could show that the GBS was caused by the administration of the vaccine. What was the basis of this stipulation to liability? Was the U.S. government taking the position that the law imposed strict liability, that

the warning had been inadequate, or that the vaccine was negligently manufactured or administered?

The court in In re Swine Flu Immunization Products Liability Litigation , 533 F. Supp. 703, 718 (D. Utah 1982), reported the following:

> During pretrial procedures . . . the controversial but crucial official statement of then-Secretary of Health, Education, and Welfare, Joseph A. Califano, Jr. was made.
>
> Secretary Califano stated that with respect to those alleging GBS, the government was adopting a new policy. Persons who contracted GBS from the swine flu vaccine, in order to receive federal compensation:
>
> [W] ill not need to prove negligence by Federal workers or others in the Swine Flu Program as required by Federal law and the law in many states. Instead claimants in most cases need only to show that they in fact developed Guillain-Barré as a result of a Swine Flu vaccination and suffered the alleged damage as a result of that condition [Statement of Secretary Califano, June 20, 1978].
>
> Secretary Califano set out two reasons for adoption of this policy:

> First, the informed consent form . . . did not warn individuals that there was a one in one hundred thousand risk that a person receiving a flu shot would contract Guillain-Barré and that one in every two million would die from the condition . . . Second, in the Swine Flu program, the Federal Government, in an unprecedented effort, actively urged millions of Americans to get flu vaccination shots and funded the nationwide campaign. Thus we have decided to provide just compensation [emphasis added] for those who contracted Guillain-Barré as a result of the Swine Flu program rather than force many individuals to prove government negligence in protracted proceedings.

The court then added: "The Justice Department had stated that this policy was 'within the intent of the Swine Flu Act and its legislative history and was appropriate under the circumstances.' (Statement of Mr. Califano reiterating the Department of Justice policy statement)" [533 F. Supp. 718].[9] These positions attributed to Secretary Califano are difficult to interpret. The simplest explanation for the decision would have been that the tort claims act (incorporated in the Swine Flu Act by reference) conferred upon the Department of Justice the authority to represent the government in the litigation, that part of the authority to represent a client includes the authority to settle or compromise claims, and that

the Department determined in light of the risks and costs of litigation that a stipulation to liability for this class of cases was a prudent litigation stance. That would require a determination, of course, that there was some risk of liability, but in light of Reyes and Givens such a determination would seem to have been quite reasonable.

The explanations attributed to Secretary Califano by the court raise a number of troubling questions. First, the Secretary said that the warning did not warn of the GBS. Does this imply that there is liability if a manufacturer fails to warn of an adverse side effect even when there is little reason to expect the side effect? If, in fact, there was no reason to warn of the side effect because there was no reason to anticipate it, then why is the failure relevant to the stipulation of liability? Second, why wasn't it sufficient to warn, as was done, of "the possibility of severe or potentially fatal reactions?" Does each possible reaction have to be separately named, or named and then described? Third, what is the relevance of the fact that the government encouraged people to get vaccinated? If that should make the government liable, then surely it should make the manufacturer liable, for the manufacturer is the one that makes the vaccination possible. And if the major point is that Secretary Califano thought there should be what in his view was "just compensation," then why was that a decision for Secretary Califano to

make?[10] Congress had specifically decided in the course of considering the statute not to change the liability rules. Hence, Secretary Califano's authority to interpret the law in this fashion remains unclear.[11]

The stipulation was limited to an admission of liability in cases where the plaintiff could prove that the GBS was caused by the vaccination. Given the fact that there is no understood causal path between the vaccination and the syndrome, that was not a trivial burden of proof. However, the suspicion about a connection surfaced early, and the Centers for Disease Control instituted an intensive surveillance system, which resulted in the halt in the program and established an apparent association between vaccination and a higher rate of GBS in the 10 weeks after vaccination. Because of this government study, which provided proof of causal connection, the Department of Justice adopted the policy of admitting liability in all cases where the plaintiff had experienced an onset of GBS within 10 weeks of the vaccination. That left for litigation those cases characterized by issues such as: (1) Did the plaintiff display symptoms of GBS within 10 weeks of vaccination? (2) What were the damages? (3) Did the vaccination cause GBS in cases in which the symptoms were not observed until more than 10 weeks after vaccination? and (4) Did the vaccination have any other adverse effects, and, if

so, was there liability for them? The reported results of this litigation are summarized in Table 6.2.[12]

TABLE 6.2

Reported Swine Flu Vaccine Cases Based Upon Manufacturers' Liability Theory.

Table 6.2 reflects the fact that the government was successful in maintaining its position in court. Almost all decisions upheld the position that the government had no liability for conditions other than GBS that appeared in the first 10 weeks after vaccination. Table 6.2 also shows the inherent unpredictability of litigation. There are six decisions that depart from this pattern. The government policy was not to settle cases that did not fit within the "GBS in 10 weeks" criterion, so these six cases should represent the only claims of the 1,600 filed that did not meet that criterion but resulted in recovery.

Overall, the cases demonstrate the phenomenon, reported by others, that a few tort plaintiffs receive a large percentage of the total recovery. This same phenomenon is also documented by Table 6.3, which breaks down total payments by size grouping. This occurs, in part, because it is only in cases with the potential for substantial damages that the incentives exist for a lawyer compensated on a contingent fee basis to make

the significant investment required to pursue a legally and technologically complex case to trial.

The six cases that fall outside the basic pattern of the swine flu claims resolution are instructive because one of the risks that concerns potential product liability defendants is the risk of the unexpected result, particularly when it may cost millions of dollars.

Also, two of the cases resulted in decisions of precedential importance that have further confounded the law in this area.

Hockett v. United States, 730 F.2d 709 (11th Cir. 1984), affirmed a verdict in favor of the plaintiff of $324,275.61. (The decision below is unreported and unavailable on Lexis™.) The plaintiff received the vaccination in October 1976. She was diagnosed as having GBS six months later, with no evidence of prior onset.

The plaintiff's proof of causation rested upon the testimony of one Dr. Eylar, described as a biochemistry professor. The court of appeals summarized his testimony as follows:

> Dr. Eylar described his work with laboratory animals and experimental allergic neuritis (EAN), a demyelinating disease generally recognized by the scientific community to be an excellent animal model for human GBS. A

> demyelinating disease attacks myelin, the component of a sheath surrounding the nerve that serves a function similar to that of insulation surrounding an electrical wire. Dr. Eylar found that EAN can be produced in animals by injection of a protein designated as P_2, when accompanied by a chemical adjuvant used to heighten the reaction. . . .
>
> Dr. Eylar tested a sample from the same lot of vaccine given to plaintiff and found that it contained chicken P_2 protein. He also tested plaintiff's blood serum and found that it contained antibodies that reacted with chicken P_2. Dr. Eylar testified that he found P_2 antibodies in the blood sera of approximately 50 percent of 30 people who had been vaccinated and contracted GBS and that no P_2 antibodies had ever been found in the blood sera of "conventional" GBS patients (those who had not received a swine flu vaccination) [730 F.2d 711].

This summary implies that Dr. Eylar did not testify that he himself or anyone else had ever looked for P_2 antibodies in the blood of GBS victims who had not received swine flu vaccine. The government countered the Eylar testimony with the testimony of Dr. Brostoff, who testified that he tested a sample of the same lot of vaccine and that it did

not contain any P_2 antibodies [sic, probably did not contain any P_2].

The government also relied upon the CDC statistical study, arguing that because the plaintiff's GBS had begun not 10 weeks but 6 months after the vaccination there was no basis for finding causation. The district court rejected the study on the following grounds:

> [T]he mere fact that the time interval in plaintiff's case was significantly longer is not dispositive. The statistics reveal a skewed bell curve relation between the vaccinations and the onset of GBS; that does not rule out the possibility of a causal relationship between the two. Dr. Daniel Bader pointed out in his deposition that the existence of a bell curve relationship doesn't mean that things that fall outside the middle part of the curve, on either side, are not associated. After ten weeks, the incidence of GBS in the vaccinated population does not fall to zero—it drops to the background incidence rate. . . .
>
> Dr. Bader's observation that one shouldn't throw away the outside margins of the bell curve just because the majority of things occur in the middle of the bell curve is well taken. We know that the overall risk of GBS in the vaccinated population is approximately four

> times greater than normal, and that some of those in the vaccinated population do in fact develop GBS more than nine or ten weeks later; we also know that not all of the cases of GBS can even be explained with associations, much less proven causes. If cases such as this were to be decided on the bell curve statistics alone, no plaintiff who contracted GBS more than nine weeks after his vaccination would ever be able to recover [730 F.2d 712] [second emphasis added].

This logic, of course, treats the failure to prove the existence of a relationship as having no bearing on proof of its absence. The district court then placed heavy emphasis on the fact that the government's experts had not tested the plaintiff's blood for P_2 antibodies [730 F.2d 712]. Therefore, they were not in a position to testify that P_2 antibodies were not present in the plaintiff's blood.

The court of appeals affirmed, placing reliance on the fact that the district court had accepted the credibility of Dr. Eylar's testimony. "Had the P2 testimony not been in the case the plaintiff would have proven only a possibility. The P_2 testimony carried plaintiff's burden, and the government's statistics and discredited attack on Dr. Eylar's P_2 testimony did not eliminate that preponderance of the evidence" [730 F.2d 713].

Judge Tjoflat concurred with a separate opinion. His concurrence is revealing about the problems of dealing with issues of scientific fact in the litigation process.

> Counsel for the government did not contemporaneously object to Dr. Eylar's qualifications when he was tendered as an expert; nor did counsel object to his testimony describing the experiments he conducted or the opinion he based on those experiments. See Fed. Rules Evid. 702 and 703. Such objections would not have been frivolous. Dr. Eylar's testimony has not been well received in other district courts. See Carter v. United States, No. G79-369, CA Slip op. (W.D.Mich. Jan. 15, 1984) (court "not persuaded" by Eylar and others); Iglarsh v. united States, No. 79C 2148, Slip op. (N.D.Ill. Dec. 9, 1983) (same); Pancho v. United States, No. C-79-0429, Slip op. (N.D.Cal.1983) (Eylar theory "speculative"); Kirby v. United States, No. 79-1805, Slip op. (D.D.C. July 16, 1982) ("other scientists have not duplicated the testing . . . his latest theories have not been published. . .. Dr. Eylar's testimony . . . has not been accepted by the scientific community"); Anger v. United States, No. 80-F-105, Slip op. (D.Colo. July 31, 1981) (Eylar's theories novel, unsupported, and not sufficiently similar to GBS). As the Middle District of Florida held in Mason v. United

> States, No. 79-614-Div-Orl-17, Slip op. (M.D.Fla. Jan. 24, 1984):
>
> Were it not for Dr. Eylar's own statements concerning the developing nature of his theories and the obvious unreliability of his testing of the relevant vaccine lot, perhaps his theories would deserve more serious thought and discussion.
>
> Moreover, even accepting all of Dr. Eylar's objective tests and test results as valid, the leap in logic from his findings to his conclusion concerning the causation of GBS in humans remains monumental.
>
> By failing to make timely objection to Dr. Eylar's qualifications and testimony, and the experiments on which his opinions were founded, the government's counsel deprived the plaintiff, and the court, from further inquiry which, for all we know, may have cast Eylar in a more credible light. Certainly, justice does not require that we now give the government, having failed in its trial strategy, a second chance to prevail.

In other words, the particular attorney appearing for the government at the Hockett trial failed at the right moment and the right time (in the ex-post view of the court of appeals) to make the point that Dr. Eylar's testimony was very far from

the scientific mainstream. The failure of the government attorney conferred upon plaintiff Hockett the benefit of a windfall recovery that no one else in her situation enjoyed.

In Sulesky v. United States, 545 F. Supp. 426 (S.D. W. Va. 1982), the court found that the "onset of the Plaintiff's GBS occurred on approximately February 1, 1977 . . . in the fourteenth week after the administration of her swine flu shot" [Id. at 428]. The court first found that the statistical studies did not disprove that the plaintiff's GBS had been caused by the vaccine because their basic method was to compare the incidence of GBS in the vaccinated population against a background rate, and "the actual background rate for GBS has not yet been established . . ." [Id. at 430], pointing to disagreement about the precise background rate. Then the court found "that the resolution of the causation issue turns on the testimony of the treating and evaluating physicians." Although the physicians had originally diagnosed the plaintiff as suffering from upper-respiratory illness, the treating physician testified at trial that she had GBS. In addition, the court heard "the compelling testimony of Dean Robert Waldman which convinced the Court . . . that this case turns upon the uncontroverted diagnosis which he made based upon his evaluation of all of the medical evidence before him in the individualized case of Kathryn Sulesky."

The court took this approach, although there is nothing in the scientific literature to suggest that there is anything about an individual patient's condition or history that enables anyone to distinguish between GBS caused by a vaccination and GBS caused by something else. A treating doctor can diagnose a condition based upon his examination of the patient at the time of examination. The notion that such a doctor also has the expertise to determine the cause of the condition is one that has been discarded in areas such as proving the efficacy of drugs. In that area of the law, a drug is not approved for use because a doctor is of the opinion that a particular patient that he treated was helped by the drug. Scientific evidence is required.

In Spencer v. United States, 569 F. Supp. 325 (W.D. Mo. 1983), the plaintiff-decedent was vaccinated on November 2, 1976, and first sought medical treatment on February 25, 1977, more than 15 weeks later. That visit to his family physician was inconclusive, and he sought no further medical treatment until his hospitalization in an acute condition the morning of March 31, 1978.

> [I]n comparatively rare instances in which a low-grade, chronic malady develops, and later apparently flares into acute GBS, the courts and the medical profession are divided in their

> conclusions as to causation. A small number of experts, including Dr. Poser [who testified for numerous plaintiffs in the swine flu cases], have recognized a related phenomenon of 'smoldering' and 'flaring' . . ." [569 F. Supp. 328].

The court then reviewed testimony of family members that the decedent had experienced "early symptoms involving the leg and hands," 569 F. Supp. 329, and concluded:

> Plaintiff's expert, Dr. Poser, is obviously well qualified and familiar with all aspects of both past and current literature on the subject. The government's expert testimony was from less highly qualified witnesses. Dr. Vernon I. Rowe, a local witness, was not unimpressive, but he appeared to testify more from knowledge of orthodox theories and materials published in past decades than from current and thorough familiarity with the applicable field. It may be argued that Dr. Poser is more of an advocate, having been retained in many of the swine flu cases, but the Court found his testimony more persuasive than that of the less-knowledgeable witnesses presented by the government.
>
> Consideration of the medical issues does not, of course, turn entirely on Dr. Poser's testimony, but a detailed review of the pertinent literature

> and material presented to the Court would not be worthwhile. This would tax not only the Court's understanding of the details of the medical debate but also its ability to articulate the debated issues in medically correct jargon. . .. The Court does emphasize, however, the usefulness of current expert testimony directed toward the facts of the case [569 F. Supp. 330].

Thus did "knowledge of orthodox theories and materials published" fall to the more speculative theories of Dr. Poser.

Gassman v. United States, No. 79-314-ORL-CIV-06 (M.D. Fla. June 27, 1984) involved a plaintiff who fell ill within 2 weeks after receiving the vaccine. She, however, suffered not from GBS but from what her neurologist diagnosed at the time of her discharge from the hospital as "probable vital encephalitis, probably with secondary transient cerebral ischemic episodes." Again the treating doctors testified that the condition was caused by the vaccination, principally on the ground that they had not been able to figure out what else caused it. The government's expert offered knowledge of the epidemiological literature, but little knowledge of the plaintiff. The court was impressed by the skill of the plaintiff's doctor in treatment, and therefore accepted his testimony on the issue of causation. Even the

government's expert, noted the court, conceded that the plaintiff's doctor had done "an excellent job" in treating the plaintiff [Id. n. 10].

Again the court made the point that a failure to prove a statistical association does not prove that the vaccine did not cause the condition. "Although these numbers [of reported cases of neurological syndromes other than GBS] are much lower than the reported cases of Guillain-Barré syndrome (more than 500 [as opposed to 24]), and the overall incidence of encephalitic disorders among those vaccinated may not have risen significantly, clearly one cannot logically exclude the possibility that certain of these reported cases were caused by the swine flu vaccine (just as one cannot logically exclude the possibility that certain reported cases of Guillain-Barré syndrome were not caused by the swine flu vaccine)."

The court then offered a theory of causation derived from the plaintiff's experts.

> It was undisputed that Mrs. Gassman's neurological syndrome included a mild polyradiculitis which produced the generalized aches and pains in her thighs beginning shortly after the vaccination. This finding by Dr. Derbenwick is significant, in the court's view, because polyradiculitis, an inflammation of the nerve roots, is characteristic in cases of

> Guillain-Barré syndrome. It would seen reasonable to conclude, therefore, in accordance with Dr. Derbenwick's testimony, that Mrs. Gassman most likely experienced an inflammatory response to the swine flu vaccine which, although idiosyncratic in that it occurred mainly in the brain, was pathogenetically related to the sort of inflammatory response, involving predominately the peripheral nervous system, experienced in cases of post-vaccinal Guillain-Barré syndrome.

This passage suggests that the court thought that polyradiculitis was a distinctive feature of post-vaccinal GBS, whereas, in fact, it is a feature of all GBS. There is no demonstrated association between polyradiculitis and the vaccine, only between GBS and the vaccine.

The issue of causation having been resolved in favor of the plaintiff, that left the issue of liability. The court first complained of the "unfairness" of the government's stipulation of liability in the GBS cases. "By rejecting in all other cases the interpretation it has given the swine flu act in Guillain-Barré syndrome cases, the government places an unfair burden, in the court's opinion, upon plaintiffs such as Mrs. Gassman who have proven that the swine flu vaccine caused a related neurological disorder. . .. During a hearing . . . the court inquired as to why the government's

Guillain-Barré syndrome policy should not collaterally or equitably estop it from claiming in other cases that, e.g., the informed consent form was adequate" [Id. n. 16]. The court then went on to find liability on the ground that the consent form was inadequate.

The issue of the adequacy of the consent form was crucial in Gassman and the two court of appeals cases yet to be discussed. It was a brief and fairly straightforward document (see Pure Politics and Impure Science by Arthur M. Silverstein6). Under the heading "Possible Vaccine Side Effects," it said (in full): "Most people will have no side effects from the vaccine. However, tenderness at the site of the shot may occur and last for several days. Some people will also have fever, chills, headache, or muscle aches within the first 48 hours." Then followed a new heading, "Special Precautions." It said, "As with any vaccine or drug, the possibility of severe or potentially fatal reactions exists." If the warning was legally adequate, this was the sentence that made it so.

The court in Gassman found the warning inadequate because it was under the wrong heading and because it did not enumerate the specific complications that constituted the "severe or potentially fatal reactions." A judgment of $90,673.95 was entered.

In Unthank v. United States, 533 F. Supp. 703 (D. Utah 1982), affirmed 732 F.2d 1517 (10th Cir. 1984), the plaintiff was vaccinated on October 18, 1976, and experienced a sudden onset of severe pain in her lower back, followed by tingling and numbness in her legs and torso 4 weeks later. She was hospitalized and diagnosed as suffering from transverse myelitis. The plaintiff's experts (not the treating doctor) testified that the vaccine caused the transverse myelitis, reaching this conclusion "by ruling out other possibilities" [533 F. Supp. 710]. The defendant's expert testified, without conducting an examination of the plaintiff, that her condition was caused by a preexisting multiple sclerosis [see the summary of the expert medical testimony, 533 F. Supp. 709-713]. The court followed the theories of Dr. Poser and found causation. That left the question of liability.

The district court first rejected the argument that the warning was inadequate, relying on its own previous decision, Bean v. United States, 533 F. Supp. 567 (D. Colo. 1980). "[T]he inclusion of potential neurologic disorders resulting from the vaccine adds nothing to the knowledge that the vaccine may lead to 'severe or potentially fatal reactions'" [Id. at 715].

The court then adopted a direct path to liability. It found that the statute required compensation for all conditions caused by the vaccine—a unique

construction of the statute. Alternatively, the court based liability upon Secretary Califano's concession of liability in the case of GBS. "The fact that GBS alone was singled out by Secretary Califano does not justify differing treatment for other 'closely related' disorders proven to have been caused by the swine flu vaccine" [533 F. Supp. 722]. The legal principles invoked were not clear. The district court then discussed at length the need for a national compensation program, citing all the authorities who have urged one since 1976. "The field of national immunology cries out for a more expeditious and fairer way of determining legitimate claims and compensating victims of the vaccination" [533 F. Supp. 726-727].

The court of appeals affirmed [732 F.2d 1517 (10th Cir. 1984)]. "We believe the trial court's ultimate conclusion that the federal government was liable must be sustained, not only on the ground which the trial court purported [sic] to give but on at least two additional theories . . ." [732 F.2d at 1519]. The court then threw a few new ideas into the liability debate. First, the Califano concession required liability because it really included transverse myelitis, because Guillain-Barré syndrome "is a broad, encompassing term which would include the transverse myelitis" [732 F.2d at 1520]. The court went on to reinterpret the trial court conclusion that the warning was adequate:

> [E]ven though the trial court expressly purported to reject a theory of inadequacy of the informed consent in this and a previous case not appealed, we think its findings belie that conclusion. Thus we conclude that the trial court's findings, as a matter of law, support the theory of inadequate informed consent.

The court then set out its own notion of what informed consent requires. "We believe that encompassed in the duty to inform a patient of all material information, substantial and significant risks, is the duty to inform not only of risks that might occur from the particular treatment in question, but also any alternative treatments and the risk of no treatment at all."

The court emphasized the importance of the setting of the swine flu program.

> A barrage of publicity aimed at overcoming the reluctance of citizens to participate included the unprecedented appearance of the President of the United States on national television to plead for a positive response. Against that background, it would be a travesty to suggest that people who hurriedly signed the standardized form presented to them were adequately informed of the risks [732 F.2d at 1521].

Based on this combination of factors, the court concluded that the government was liable under the standard of section 402 of the Restatement .[3]

Petty v. United States resulted in three reported decisions at 536 F. Supp. 860 (N.D. Iowa 1980), 679 F.2d 719 (8th Cir. 1982), and 740 F.2d 1428 (8th Cir. 1984), and one unreported decision, N.D. Iowa, No. C 78-4083, March 31, 1983. The plaintiff experienced severe symptoms resulting in hospitalization 8 days after his vaccination. The treating doctor diagnosed this condition as "serum sickness-like reaction to a foreign protein." The treating doctor testified that this condition was caused by the vaccination. The government presented no live medical expert, relying on depositions and exhibits.

> At trial, Dr. Hyden [the treating doctor] testified that in his medical opinion the swine flu shot caused plaintiff's severe serum sickness, which in turn caused plaintiff's congestive heart failure. Dr. Hyden testified that the vaccine was a foreign protein which caused an immune complex to form, setting up an antibody-antigen reaction, which created an inflammatory condition throughout the body. Dr. Hyden testified that there was no evidence that plaintiff received any other foreign protein besides the swine flu vaccine.

> . . . The Court is of the opinion that because Dr. Hyden was plaintiff's personal physician and examined plaintiff throughout the relevant times involved in this matter, that his testimony is the more convincing in this matter" [536 F. Supp. 871].

The government had not stipulated to liability for "a serum-like sickness," so the problem of liability remained. The court found liability on the ground that the information form failed to meet the standards of the swine flu statute, although there was nothing in the statute to suggest that a failure to meet its requirements would result in liability of the government. Of the form, the court said:

> [B]ecause of the "hard sell" campaign, the Government has a responsibility to protect and help those citizens who volunteer for the vaccination and subsequently become ill as a result thereof. The Government should not hide behind the four corners of a complicated registration form (now argued to be a consent form) and thereby deny and avoid liability [536 F. Supp. 870].

The court of appeals reversed. The district court erred, it said, because the district court found that the liability arose under the swine flu act itself. Rather, said the court of appeals, liability was to be controlled by state law. Judge Bright, writing for

the court, reversed and remanded the case to the district court for findings under the standards of Iowa's informed consent law [679 F.2d 727-729].

On remand, the district court found that the information form failed to meet the standards of Iowa law. The court of appeals affirmed. It affirmed because the Iowa statute requires disclosure of the known risks "of death, brain damage, quadriplegia, paraplegia, the loss or loss of function of any organ or limb, or disfiguring scars associated with such procedure or procedures, with the probability of each such risk if reasonably determinable." The swine flu information form had said only "the possibility of severe or potentially fatal reactions exists," failing to enumerate in detail the potential reactions and the risks of each. Because the form had failed to warn Petty of the risk of "serum sickness," the manufacturer would have been liable under Iowa law. The court placed particular emphasis on the government's "unprecedented promotional campaign," and on the Califano statement explaining the stipulation to liability for GBS. The court concluded:

> [T]here is a public policy consideration favoring allocating the risk of loss among the distributors who can then absorb the cost or distribute it more easily. See Reyes v. Wyeth Laboratories, 498 F.2d 1264 n. 57 (recognizing

> view that burden should be borne by the government because of the societal benefit bestowed by individuals receiving the vaccine).

The court went on to make it clear that its holding was not limited to the government under the special conditions of the swine flu statute and vaccination program. It was a holding that the manufacturer would have been liable under Iowa law if it had not been for the swine flu act. "Under Iowa law, Merrill-National would have been held strictly liable for its distribution of a defective product which proximately caused Petty's injuries" [740 F.2d 1441].

Judge Bright dissented. His dissent in full [740 F.2d 1441-1442] was as follows:

> I dissent. I believe that the warning Petty received before being vaccinated was adequate. Therefore the United States is not liable under either the negligence or the strict liability theory propounded by the majority.
>
> Even assuming that serum sickness was a foreseeable risk attending swine flu vaccination, I think vaccinees received adequate warning. The information form Petty read stated that, '[A] s with any vaccine or drug, the possibility of severe or potentially fatal reactions exists.' I do not think that a more specific warning, either describing the

symptoms of serum sickness or mentioning that condition by name, would have served any useful purpose in the context of this mass inoculation program. Vaccinees were warned that a small risk of severe or fatal reaction accompanied receiving the vaccine. The majority does not dispute the minimal nature of the risk, but maintains that the warning should have itemized the potential adverse reactions from the vaccine. Yet, the warning of the "possibility of severe or potentially fatal reactions" more directly, completely, and graphically describes for the lay person the hazards of receiving the vaccine than a specific itemization of adverse effects. I agree with the government and with most of the courts that have considered the issue, see supra N.6., that a detailed catalogue of every serious complication that might befall a vaccinee would have been counter productive, serving to confuse or needlessly alarm potential vaccinees without giving them any more information necessary to the making of an informed decision. I therefore reject the majority's conclusion that the failure to specify the potential adverse effects of the vaccine makes the warning inadequate when, as here, the warning aptly apprised vaccinees of the overall possibility of harm.

> Because the warning was adequate, the government was not negligent. Nor is the government strictly liable under the theory that the unavoidably unsafe vaccine was rendered defective or unreasonably dangerous by the failure to give a more detailed warning.
>
> I think the practical consequence of the court's decision is to impose so stringent a warning requirement as likely to render any future mass inoculation program infeasible, no matter how desirable.
>
> Accordingly, I dissent.

The mixture of liability theories put forward in Petty is comprehensive. Most significantly, the court emphasizes the primary importance of the state informed consent law (as did the district court in Gassman). The informed consent statutes have been passed in many states to provide a clear procedure by which a doctor, in the context of a direct doctor-patient relationship, can obtain consent to a medical procedure that may constitute battery. They set standards for genuine consent, requiring full disclosure. The statutes deal neither with malpractice nor product liability, nor with mass health care procedures such as a public vaccination program in which the vaccinee is not a "patient," in the usual sense, of anyone. The implication in the Petty case is that those

statutes now set the standard for vaccination and that failure to meet the standard results in liability for any resulting damages, apparently liability imposed on all participants in the vaccination process. This, combined with the favorable citation of a portion of the Reyes opinion, makes Petty a notable development in the area.

What conclusions can be drawn from the swine flu litigation about the risks of a manufacturer's liability? That question depends upon the extent to which litigation occurred under circumstances similar to those that would be faced by a vaccine manufacturer. There are a number of features of the swine flu litigation that are different.

First, the government approach to the problem of defending tort claims is different from that of private parties. Defendants in tort suits with large claims make use of substantially more lawyers and legal resources than does the government. The government, in accordance with its usual tort claims procedures, assigned defense of these numerous tort claims to a small law office, assisted on an intermittent and erratic basis by generalist lawyers with many other duties from the offices of the various U.S. attorneys. This shows up in many aspects of the government defense, particularly its heavy reliance on documentary submittals and expert witnesses who had not examined the particular plaintiff. Indeed, the most

straightforward explanation for the government's stipulation to liability in the "GBS within 10 weeks" cases may be that the stipulation was necessary to make the cases manageable—the stipulation left for litigation only those cases in which the government had a straightforward defense that could be submitted largely on paper.

The second important distinguishing feature is that plaintiffs under the tort claims act (incorporated by reference into the swine flu statute) do not have a right to trial by jury. All swine flu cases were tried in bench trials, in which the judges alone heard the evidence and made the findings of fact. It is widely believed that juries increase the unpredictability of outcomes, and tend to be more sympathetic to plaintiffs.

A third important distinguishing feature is that all claims under the tort claims act must be brought in the federal district courts. Many cases against a vaccine manufacturer would be brought in state courts. The differences among state courts would introduce another source of variance in such litigation. The federal district judges have lifetime tenure and historically have received premium salaries.

Although there are distinguished state judiciaries, in a "worst case" scenario a vaccine manufacturer from a distant state would face trial in a state

court located in the hometown of the plaintiff, and controlled by a judge subject to frequent elections before that hometown electorate. The plaintiff's case is likely to be supported by "expert" testimony from the treating doctor, perhaps well known and respected in the community.

Fourth, the swine flu program was accompanied by much more intense publicity and news coverage than regular, ongoing vaccination programs. That news coverage may have made many more people aware of the possible connection between the vaccine and GBS, it may have made more recipients consider filing claims, and it may have made doctors more sensitive to the possibility of a GBS diagnosis in persons who received a vaccination. [13]

Fifth, there is a difference between the swine flu claims and the claims that would confront a manufacturer of pediatric vaccines in an analogous situation. The swine flu vaccination program was focused on adults and particularly on older persons more likely to incur serious complications from the disease. This affected the litigation in two ways. First, older people have shorter life expectancies and any element of damages calculated over life expectancy is reduced. Second, in many states, a statute of limitations does not run against a minor. Thus, a 1-year-old who suffers injuries as the result of a

vaccination may have more than 20 years before any claim is cut off by the passage of time. This makes it more difficult for the manufacturer in the pediatric case to estimate the amount of potential outstanding claims.

Sixth, it is hard to overlook the fact that the defendant in the swine flu litigation was the ultimate "deep pocket"—the government of the United States. The judges in the swine flu litigation were not confronted by a vaccine producer whose solvency and future ability to provide vaccines would be critically affected by the outcome of the litigation. The defendant, instead, was the United States, which would hardly notice the difference. This fact affected not only the reactions of the judges, but the decision to stipulate to liability, the resources dedicated by the Department of Justice to the defense of the cases, and the style of argument that could be made to the courts.[14] Indeed, in retrospect it may become clear that the major mistake of the swine flu statute was to place the defense in the hands of a bureaucracy without any long-term stake in the controlling rules.

What can be concluded from the swine flu litigation?

First, the professionals close to the plight of a particular victim find it hard not to support the

merits of the claim. The treating doctor, perhaps not unhappy to direct a potential malpractice claim elsewhere, provides supportive testimony. Trial judges, confronted by the hardship the plaintiff has experienced, are tempted to adopt new interpretations of the law.

Second, doctors and lawyers prefer to treat the issues as the other profession's problem. The lawyers and judges involved with a particular case focus their discussion on issues of medical fact, and ostensibly base their decisions on which medical expert's testimony is more believable. Many doctors, on the other hand, think the law should provide a solution to the problem: some are unwilling to testify, and others fail to differentiate in their testimony between established effects and ideas outside the scientific mainstream.

Third, the reaction of the courts who found liability to Secretary Califano's stipulation vividly illustrates the problem facing the manufacturers. One might argue that the manufacturers have a simple solution to the liability burden and the costs of litigation. Instead of strongly resisting liability, as they have, they could simply concede liability and settle, recovering the costs of their settlements in higher vaccine prices and avoiding all the costs of extended and unproductive disputes over issues of esoteric medical fact. That is what Secretary

Califano basically did when he supported the government's stipulation of liability for GBS, but a concession of liability builds upon itself. The pressure to treat like claims alike generates an ever-expanding pool of liability if the original concession of liability is based upon no identifiable theory or principle.

Finally, the Unthank and petty opinions seriously erode any confidence that in the future a defendant faced with substantial claims could duplicate the government's basic success. In spite of the fact that the government won almost all of the cases it contested, Unthank and Petty are court of appeals' opinions strongly favoring liability. If they had been handed down early in the litigation, they might have been viewed as setting a precedent in favor of recovery for the plaintiff.

Recently, a few other cases relating to liability for the manufacture of vaccines have been reported. One is a flu vaccine case, involving an administration during the same fateful fall of 1976, but of a monovalent vaccine (Victoria A strain only) administered in a doctor's office and thus not falling within the swine flu statute [Stanback v. Parke-Davis & Co., 657 F.2d 642 (4th Cir. 1981]. The plaintiff had contracted GBS, and the defendant had not warned of the risk of GBS. The doctor testified that it was his practice not to warn patients of the type of risk, and that, although he

was aware of a possible risk of GBS in the fall of 1976, he did not warn the patient and would not have warned the patient even if the package insert had been different. The district court granted a summary judgment to the defendant for failure of the plaintiff to prove a causal connection between the manufacturer's failure to warn and the plaintiff's injury. The court of appeals affirmed.

There also have been additional polio cases. The most notable is Schindler v. Lederle Labs., 725 F.2d 1036 (6th Cir. 1983), which affirmed a judgment in favor of the manufacturer on the grounds that its package insert provided adequate warning of the risk of contracting polio. The doctor had administered the vaccine under conditions that are contraindicated. Dunn v. Lederle Laboratories, 121 Mich. App. 73 (1982), also affirmed a verdict for the defendant. In Loge v. United States, 662 F.2d 1268 (8th Cir. 1981), the district court had dismissed the complaint against the United States. The court of appeals reversed on the grounds that the plaintiff's allegations that the United States had approved the vaccine in violation of its own regulations were sufficient to state a cause of action. In Fraley v. American Cyanamid Co., 570 F. Supp. 497 (D. Colo. 1983), the court held that the defendant was collaterally estopped from contesting the adequacy of a warning accompanying a dose administered in 1971 and identical to the warning in Givens.

There are two reported decisions relating to DTP. Walton v. Charles Pfizer & Co., Inc., 590 P.2d 1190 (Okla. 1979), affirmed a verdict (of $2,000, a statutory limit) against the City of Tulsa for the 1966 administration of pediatric vaccines (probably DTP, but not clear). Morris v. Parke-Davis & Co., 573 F. Supp. 1324 (C.D. Cal. 1983), an ongoing case, involves a claim based on a DTP vaccination given in 1965. The plaintiff seeks actual and punitive damages. All manufacturers who produced DTP at the time are joined as defendants because the source of the DTP is not known. A motion to strike the claim for punitive damages was denied by the court.

Caron v. United States, 548 F.2d 366 (lst Cir. 1976), affirmed a judgment of $705,606 for immunization of a 4-month-old baby with DTP, oral polio vaccine, and typhoid vaccine leading to convulsions, grand real seizures, and permanent mental retardation. The typhoid given was an adult dose. The claim was brought 10 years after the vaccination on the theory that the typhoid vaccine was administered negligently, both because the dosage was improper and because in the absence of a special risk of typhoid, it should not be given to a baby.

Lemar v. United States, 580 F. Supp. 37 (W.D. Tenn. 1984), dismissed a suit against the government for encouraging pediatric vaccination

without warning of possible adverse consequences.

Calabrese v. Trenton State College, 162 N.J. Super. 145 (1978), held the manufacturer, distributor, and seller not liable for damages allegedly caused by rabies vaccine, but denied the physician's motion for summary judgment because a fact issue existed as to whether he had disclosed possible adverse side effects. Hitchcock v. United States, 479 F. Supp. 65 (D.D.C. 1979), awarded $519,051 against the united States to the wife of a foreign service officer who was given anti-rabies vaccine as "pre-exposure prophylaxis" by government doctors prior to a foreign posting. The liability was based on the failure of the government to warn of the benefits and risks of the vaccine.

Liability for injury resulting from vaccination is a matter of state law. Thus, there are potentially 51 different sets of rules on this subject in the United States. No vaccine-related product liability litigation resulting in a publicly available opinion has occurred in most jurisdictions, and in many there has been no liability litigation related to drugs. Although it is possible by extrapolation from other product liability cases to predict how these states would handle vaccine cases, any such extrapolation might prove inaccurate.

The subject of product liability has become increasingly controversial over the past few years. Manufacturers have complained about the costs, the unpredictability of the law, and the unavailability and cost of insurance. A majority of states have enacted statutes relating to some aspect of product liability actions.[1] Legislation at the federal level has been considered extensively, but has not been passed. [2]

A manufacturer who produces and sells a defective vaccine that creates a risk of significant injury to the recipient is liable to any person injured by that defect under the principles stated in section 402A of the Restatement of Torts 2d.[3] This is

thought to be the law in every American jurisdiction. The key portion of section 402A states:

Cases holding manufacturers or others liable for allegedly defective vaccines administered to humans are very rare.[4] The reported cases are:

A manufacturer is not liable for harm caused by a nondefective product due to its inherent or unavoidable dangerousness. Thus, if a properly manufactured vaccine will cause harmful side effects in some portion of the recipient population, the manufacturer of the vaccine is not liable for those side effects. This principle is the subject of comment k to section 402A. The qualification "proper warning is given" is critically important and a manufacturer will be liable if he markets a drug or vaccine with known risks and fails to warn of them, and it can be shown that the recipient would not have taken the drug or vaccine had he known of the risks. An example of this situation involving polio vaccine is Davis v. Wyeth Laboratories, Inc., 399 F.2d 121 (9th Cir. 1968). In that case, Sabin polio vaccine had been administered to the plaintiff, a 39-year-old male, in a mass immunization campaign conducted with the assistance of a sales representative of the defendant. The plaintiff had developed polio. The Surgeon General had recommended that mass programs using Sabin Type III be limited to children because of the risk of transmission of the disease to adult recipients. The mass immunization campaign involved in the case, however, included adults. The defendant's promotional materials failed to warn of the risk to adults, although the package insert did. The pharmacist who dispensed the oral vaccine did not read the package insert and did not warn the plaintiff of the risk.

The duty to warn arises from the notion that the failure to inform consumers of dangerous aspects of a product may render the product unreasonably dangerous. For instance, if an electric device operated in a certain way will blow up and cause harm to bystanders, a failure to warn the consumer of this risk so that he can avoid the harm will make the manufacturer liable for the harm. Or in the Davis case, if the plaintiff had been warned of the risk, he could have chosen not to take the vaccine, as was then recommended in the case of adults.

If a product is not sold directly to the public, but is distributed through intermediaries who can be expected to know about the product and its risks and to be responsible for informing the ultimate consumer on its proper use, then the manufacturer does not have a duty to warn the public (although it does have a duty to warn the intermediaries of risks not known to them). Prior to Davis, this exception to the duty to warn was thought to be applicable to drugs and vaccines distributed through health

professionals (as contrasted with over-the-counter drugs, for which the manufacturer does have a duty to warn the user).
In Davis, the defendant manufacturer argued that its duty to warn had been satisfied by the fact that it had informed the county medical society sponsoring the mass immunization program of the risk that a vaccinee might develop polio. The court rejected this argument in the context of the mass immunization program:
The Missouri Supreme Court reached the same result in Stahlheber v. American Cyanamid Co., 451 S.W. 2d 48 (Mo. 1970), involving another mass administration of oral polio vaccine. Recipients included a 41-year-old adult who contracted polio. The manufacturer had not warned that the vaccine should not be administered to adults other than those facing special risks of contracting the disease.
The law summarized here appears to lead to the conclusion that a vaccine manufacturer need not fear liability from injuries resulting from the administration of the vaccine if (1) the vaccine is correctly manufactured in accordance with regulatory standards and prudent manufacturing practices and (2) the manufacturer has taken reasonable steps to ensure that the recipient of the vaccine will be warned of possible side effects. Merrill's comprehensive 1973 survey of the law of liability for all drugs concludes: "If a manufacturer adequately warns physicians [or, in the case of mass-administered vaccines, the recipient] about a drug's foreseeable adverse effects, he will escape liability unless the plaintiff can show that his injury was caused by some impurity or resulted from an unreasonably dangerous design" [Merrill, "Compensation for Prescription Drug Injuries," 59 Va. L. Rev. 1, 49 (1973)].
In two decisions, the Fifth Circuit Court of Appeals, nominally applying the law first of Texas and then of Florida, imposed liability in circumstances that would not be covered by these rules. Both decisions have been subject to criticism,[5] and it is not clear how many courts would follow them today.
The first and most important of these decisions is Reyes v. Wyeth Laboratories, 498 F.2d 1264 (5th Cir. 1974). That case involved a young girl to whom Sabin oral polio vaccine was administered at a county department of health clinic in Mission, Texas. She developed polio, and her parents sued on her behalf for damages.
The manufacturer's main line of defense at the trial was that the vaccine was not the cause of the disease. Experts testified that virus isolated from a specimen of the child's stool taken on the day after she was admitted to the hospital was "probably wild," i.e., naturally occurring and not the same strain as the vaccine [498 F.2d 1270-1271]. The jury found, however, that the

vaccine was the cause of the child's disease, and found for the plaintiff under the district court' s instructions.

The critical issues decided by the court of appeals were two. First, had the manufacturer discharged its duty to warn? And second, was the failure to warn sufficiently connected to the occurrence of the child's disease to support liability? Judge Wisdom wrote a unique opinion strongly affirming the judgment for the plaintiff.

In Reyes, Wyeth had played no role in the vaccination program, perhaps because it had changed procedures after its loss in Davis. It had simply shipped the vaccine in response to an order from the Texas State Department of Health. The package insert warned of the risk, and Wyeth argued that it was all that was required to discharge its responsibility. The procedures, forms, and warnings actually used in the immunization program were the responsibility of the Texas Department of Health, to which it had quite properly shipped the vaccine. Judge Wisdom answered this argument without really meeting it. Wyeth knew that the vaccine was customarily administered in mass administration programs staffed at least in part by volunteers and nonprofessionals. Knowing that each recipient would not have individualized medical advice, it "was required to warn foreseeable users, or see that the Texas Department of Health warned them" [498 F.2d 1277]. Exactly how Wyeth was to provide these warnings at distant clinics with which it had no connection, or why Wyeth was not entitled to rely upon the Texas Department of Public Health to do its job properly, Judge Wisdom did not clarify. To the argument that Davis was inapplicable because in that case Wyeth had actively participated in the program, Judge Wisdom observed that "the present controversy, however it differs from Davis factually, invites application of the Davis principles, and the conclusion that Wyeth was under a duty to warn . . . [the recipient's parents]" [498 F.2d 1277].

The second issue confronting the court of appeals was whether there was sufficient connection between the failure to warn and the injury to the child. There was no demonstration that had the child's mother been warned of the minute incidence of polio from the vaccine she would have decided not to have her child immunized. This possibility would seem to have been extremely unlikely because polio was occurring in the area. The court first looked to Texas cases for the following rule:

In the normal product liability case, this rule makes sense. If the consumer is warned of a danger, he or she can take steps to avoid it.

However, what was the parent confronted with the warning that Sabin polio vaccine might cause polio to do? Forgo vaccination and incur the risk of contracting polio naturally? No, said the court, take Salk killed-virus vaccine first.

This aspect of the decision has been vigorously criticized by Franklin and Mais in "Tort Law and Mass Immunization Programs: Lessons from the Polio and Flu Episodes," 65 Calif. L. Rev. 754 (1977):

The Oklahoma Supreme Court did not follow Reyes in Cunninqham v. Pfizer & Co., 532 P.2d 1377 (Okla. 1976), because the evidence in that case showed that at the time of the plaintiff's vaccination in 1963, Oklahoma was in "an epidemic state." The court reversed a verdict for plaintiff and remanded, requiring the plaintiff to show that if the warning had been given, he would not have been vaccinated.

In Reyes, the court imposed liability on the manufacturer for failing to warn the recipient directly of a risk of contracting poliomyelitis from the vaccine even though the warning accompanied the vaccine and would have been very unlikely to change the conduct of the recipient, or to have prevented the child from contracting polio.

Judge Wisdom's opinion closed with an extraordinary section in which, in response to an amicus brief filed by the American Academy of Pediatrics and the Conference of State and Territorial Epidemiologists opposing liability, he took on both medical practice and the Congress. To the doctors he said:

Givens v. Lederle, 556 F.2d 1341 (5th Cir. 1977), was a further extension of Reyes. Givens, like Reyes, involved Sabin oral polio vaccine. However, unlike Reyes, the vaccine had not been administered in a county health clinic, but by a pediatrician in his office. The victim was not the child but the child's mother, who had apparently contracted the polio from the vaccinated child. The pediatrician had given no warning of this or any other risk, although the risk was disclosed in the package insert.

The case had concluded in a jury verdict for the defendant in the Middle District of Florida (then part of the Fifth Circuit), on the grounds that the vaccine could not have caused the polio [556 F.2d 1343]. Reyes was handed down before the entry of judgment, however, and the trial judge granted a new trial in light of Reyes. On retrial, the verdict was for the plaintiff, and the defendant appealed.

A remarkable aspect of the Givens opinion is that the court did not rely on Florida law to support its judgment. In Reyes, the court had gone to great lengths to highlight a line of Texas cases concerning possible allergic reactions to a drug. In these cases, it was held unnecessary to warn recipients of possible allergic reactions if the number of people affected was not significant. It might appear that because the number of people who get polio from the vaccine is minute, this line of cases would suggest that no warning of the risk was necessary under the controlling Texas law. The court, however, distinguished between the cases on the ground that an allergic

reaction affected only a small group of recipients, while all recipients of the polio vaccine were apparently subject to the minute risk of vaccine-induced polio [498 F.2d 1278-1279]. In contrast, the Givens opinion made no reference to Florida law. It simply followed Reyes—ostensibly applying Texas law—and treated it as Florida law.

"[A] proposed major distinction is that a county health clinic administered the vaccine in Reyes, whereas a private pediatrician did so here," said the court [556 F.2d 1344]. The manufacturer apparently was relying on the long line of authority indicating that the duty to warn in the case of prescription medicines prescribed by doctors belongs to the doctor, not the manufacturer. The pediatrician in Givens had an established doctor-patient relationship, was a trained and licensed professional, and had determined that the patient should receive the vaccine as part of her health care. This was not a case of an unadvised patient responding to the entreaties of a mass vaccination campaign.

The court responded to this argument as follows: "The administration of the vaccine by a public health nurse in Reyes is as close to the instant situation as it is to the Davis mass inoculation" [Ibid]. Testimony by the plaintiff's physician that". . . the administration in his office 'really doesn't differ' from that of the Public Health Center; 'not in the administration at all'" emphasized the similarity to the Reyes situation, rather than his role as an informed intermediary [556 F.2d 1345].

The defendant also argued that its warning to the doctor was adequate, and should protect it from liability. The warning stated:

Of this warning, the plaintiff's doctor testified: "[T]he wording of the inserts states that it is a safe and effective means of immunizing the population and that the risk, if it exists, is no more than 1 in 3 million. I felt this was a very nebulous way of putting it, . . . and I did not feel there was sufficient evidence or warning to warn Mrs. Givens about this" [556 F.2d 1345]. The court held that this was sufficient evidence to sustain a jury verdict that the warning was inadequate.

Thus, the court of appeals in Givens treated as factual questions for the jury whether or not the vaccine caused the polio, and whether or not the warning was adequate in light of the actual degree of risk that the vaccine might cause the polio. This situation, in which a jury confronted with a grievously ill child or adult (who has no other potential source of financial support or redress) is asked to judge retrospectively the actions of the manufacturer, the regulatory authorities, and those responsible for the public health, tends toward a biased outcome in favor of the plaintiff.

Although Reyes and Givens were decided in conformity with the verbal formulas of previous product liability law, the

skeptical reader might easily conclude that they really stood for a different and simpler rule: the manufacturer is liable for all damages caused by a vaccine. Just such a rule for all prescription drugs had been advocated in Merrill's "Compensation for Prescription Drug Injuries," 59 Va. L. Rev. 1 (1973), and that article was cited by Judge Wisdom in the policy discussion in the Reyes opinion, 498 F.2d 1294 n. 57. Even though the Fifth Circuit had no power to make law on this point for Florida and Texas (the Texas or Florida courts could repudiate these decisions at any time if they had a case in point and chose to do so), the skeptical lawyer advising a manufacturer could take as the message from the judges: manufacturers are going to pay, one way or the other.